Higher Order Homogeneous ODEs

- **Case 2: Repeated Real Roots.**
- Suppose a real root r occurs with multiplicity m.
- Then it contributes m linearly independent solutions: $e^{rx},\ xe^{rx},\ x^2e^{rx},\ \ldots,\ x^{m-1}e^{rx}$.
- These arise from repeated application of reduction of order to the repeated root.
- If r has multiplicity m, the corresponding part of y is $(c_1 + c_2x + \cdots + c_mx^{m-1})e^{rx}$.



Higher Order Homogeneous ODEs

- **Case 3: Complex Roots.**
- Since coefficients are real, complex roots occur in conjugate pairs $\alpha \pm i\beta$.
- Each simple conjugate pair contributes two real solutions:

  $e^{\alpha x}\cos\beta x, \qquad e^{\alpha x}\sin\beta x.$
- **If repeated** with multiplicity m, the pair contributes $2m$ solutions:

  $x^k e^{\alpha x}\cos\beta x, \quad x^k e^{\alpha x}\sin\beta x, \qquad k = 0, 1, \ldots, m-1.$
- This mirrors the repeated real root case, applied to each conjugate pair.



Higher Order Homogeneous ODEs

- **Solution Space and Fundamental Set.**
- The set of all solutions of the homogeneous n-th order equation forms an n-dimensional vector space.
- A set of n linearly independent solutions $\{y_1, \ldots, y_n\}$ is a **fundamental set**.
- Independence is verified using the (generalized) **Wronskian**:

$$W(y_1, \ldots, y_n)(x) = \det \begin{pmatrix} y_1 & \cdots & y_n \\ y_1' & \cdots & y_n' \\ \vdots & & \vdots \\ y_1^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}$$

- $\{y_1, \ldots, y_n\}$ is fundamental iff $W \neq 0$ at some (equivalently every) point of I.
- General solution: $y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$.



Higher Order Homogeneous ODEs

- **Step-by-step procedure.**

1. Write down the auxiliary equation from the coefficients.
2. Factor / solve the polynomial to find all roots and their multiplicities.
3. For each distinct real root: include $e^{rx}, xe^{rx}, \ldots$ up to its multiplicity.
4. For each conjugate complex pair: include $e^{\alpha x}\cos\beta x, e^{\alpha x}\sin\beta x, \ldots$ up to its multiplicity.
5. Combine all these basis solutions with arbitrary constants to form the general solution.
6. If initial conditions are given, differentiate and solve the resulting linear system for the constants.



Problem 1

- Solve $y''' - 6y'' + 11y' - 6y = 0$.
- Auxiliary equation: $r^3 - 6r^2 + 11r - 6 = 0$.
- Testing $r = 1$: $1 - 6 + 11 - 6 = 0$ ✓, so $(r - 1)$ is a factor.
- Factoring: $(r - 1)(r^2 - 5r + 6) = (r - 1)(r - 2)(r - 3) = 0$.
- Roots: $r = 1, 2, 3$ (distinct real).
- The general solution is given by

$$\boxed{y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}}$$



Problem 2

- Solve $y''' - 3y'' + 3y' - y = 0$.
- Auxiliary equation: $r^3 - 3r^2 + 3r - 1 = 0$.
- This factors as $(r-1)^3 = 0$.
- Root: $r = 1$ with multiplicity 3.
- Basis solutions: e^x, xe^x, x^2e^x.
- The general solution is given by

$$\boxed{y = (c_1 + c_2x + c_3x^2)e^x}$$



Problem 3

- Solve $y''' - y'' + y' - y = 0$.
- Auxiliary equation: $r^3 - r^2 + r - 1 = 0$.
- Factor by grouping: $r^2(r-1) + (r-1) = (r-1)(r^2+1) = 0$.
- Roots: $r = 1$ (real), $r = \pm i$ (complex, $\alpha = 0, \beta = 1$).
- Basis solutions: $e^x, \cos x, \sin x$.
- The general solution is given by

$$\boxed{y = c_1 e^x + c_2 \cos x + c_3 \sin x}$$



Problem 4

- Solve $y^{(4)} - y = 0$.
- Auxiliary equation: $r^4 - 1 = 0$.
- Factor: $(r^2 - 1)(r^2 + 1) = (r - 1)(r + 1)(r - i)(r + i) = 0$.
- Roots: $r = 1, -1$ (real), $r = \pm i$ (complex, $\alpha = 0, \beta = 1$).
- Basis solutions: $e^x, e^{-x}, \cos x, \sin x$.
- The general solution is given by

$$\boxed{y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x}$$



Problem 5

- Solve $y^{(4)} + 4y = 0$.
- Auxiliary equation: $r^4 + 4 = 0 \;\Rightarrow\; r^4 = -4$.
- Fourth roots of -4: $r = \sqrt{2}\, e^{i(\pi+2k\pi)/4}$, $k = 0, 1, 2, 3$.
- These give $r = 1 + i,\ -1 + i,\ 1 - i,\ -1 - i$ (two conjugate pairs).
- Basis solutions: $e^x \cos x, e^x \sin x, e^{-x} \cos x, e^{-x} \sin x$.
- The general solution is given by

$$\boxed{y = e^x(c_1 \cos x + c_2 \sin x) + e^{-x}(c_3 \cos x + c_4 \sin x)}$$



Problem 6

- Solve $y^{(4)} - 2y'' + y = 0$.
- Auxiliary equation: $r^4 - 2r^2 + 1 = 0 \Rightarrow (r^2 - 1)^2 = 0$.
- So $r^2 = 1$ is a double root $\Rightarrow$ $r = 1, 1, -1, -1$.
- Each of $r = 1$ and $r = -1$ has multiplicity 2.
- Basis solutions: $e^x, xe^x, e^{-x}, xe^{-x}$.
- The general solution is given by

$$\boxed{y = (c_1 + c_2 x)e^x + (c_3 + c_4 x)e^{-x}}$$



Problem 7

- Solve the IVP $y''' + 3y'' + 3y' + y = 0$, $y(0) = 1,\ y'(0) = -3,\ y''(0) = 11.$
- Auxiliary equation: $r^3 + 3r^2 + 3r + 1 = 0 \ \Rightarrow\ (r+1)^3 = 0.$
- Root: $r = -1$ with multiplicity 3.
- General solution: $y = (c_1 + c_2 x + c_3 x^2)e^{-x}.$
- Now $y' = [-c_1 + c_2 + (2c_3 - c_2)x - c_3 x^2]e^{-x}.$
- Then $y'' = [c_1 - 2c_2 + 2c_3 + (c_2 - 4c_3)x + c_3 x^2]e^{-x}.$
- Apply initial conditions:
- $y(0) = c_1 = 1.$
- $y'(0) = -c_1 + c_2 = -3 \ \Rightarrow\ c_2 = -2.$
- $y''(0) = c_1 - 2c_2 + 2c_3 = 11 \ \Rightarrow\ 1 + 4 + 2c_3 = 11 \ \Rightarrow\ c_3 = 3.$
- So $c_1 = 1,\ c_2 = -2,\ c_3 = 3.$

$$\boxed{y = (1 - 2x + 3x^2)e^{-x}}$$



Undetermined Coefficients (Higher Order)

- **Goal.** Find y_p for $a_n y^{(n)} + \cdots + a_1 y' + a_0 y = R(x)$, constant coefficients.
- Same restriction as before: $R(x)$ must be polynomial, exponential, sine/cosine, or products/sums of these.
- **Standard trial forms** (no duplication with y_c):
- $R(x) = P_m(x)$ (degree m) $\Rightarrow$ $y_p = A_m x^m + \cdots + A_1 x + A_0$ **(0 is not root of AE)**
- $R(x) = ae^{kx}$ $\Rightarrow$ $y_p = Ae^{kx}$ **(k is not root of AE)**
- $R(x) = a\cos\beta x$ or $a\sin\beta x$ $\Rightarrow$ $y_p = A\cos\beta x + B\sin\beta x$ **($i\beta$ is not root of AE)**
- $R(x) = P_m(x)e^{kx}$ $\Rightarrow$ $y_p = (A_m x^m + \cdots + A_0)e^{kx}$ **(k is not root of AE)**
- $R(x) = e^{kx}(a\cos\beta x + b\sin\beta x)$ $\Rightarrow$ $y_p = e^{kx}(A\cos\beta x + B\sin\beta x)$ **($\alpha + i\beta$ is not root of AE)**



Undetermined Coefficients (Higher Order)

- **Modification Rule (crucial difference from 2nd order).**
- Let s be the **multiplicity** of the matching root (k, or $k \pm i\beta$) in the auxiliary equation.
- In higher order equations, s can be $2, 3$, or more (not just 0 or 1 as is typical for 2nd order).
- **Rule:** multiply the standard trial form by x^s.
- If $s = 0$: use the standard trial form directly.
- If $s = 1, 2, 3, \ldots$: multiply the entire trial by x^s to avoid duplicating y_c.
- Example: a root of multiplicity 3 matching e^{kx} forces trial $y_p = Ax^3e^{kx}$.
- For polynomial forcing matching root 0 with multiplicity s: trial multiplies x^s into the polynomial.
- For product forcing $P_m(x)e^{kx}$ with duplication: trial is $x^s(A_mx^m + \cdots + A_0)e^{kx}$.
- Superposition still applies: split $R = R_1 + R_2$, solve separately, and add.



Useful trick:

- For forcing e^{kx} matching a root k of multiplicity s, substitute $y = e^{kx}u(x)$.
- The equation reduces to $u^{(s)} \cdot (\text{const}) = 1$ (roughly), so u_p is a degree-s monomial in x.
- This avoids differentiating $x^s e^{kx}$ directly multiple times. (See Problem-3)



Undetermined Coefficients (Higher Order)

- **Step-by-step procedure.**

1. Consider the homogeneous DE and put $y = e^r x$.
2. Solve the auxiliary equation; list all roots and their multiplicities.
3. Write the complementary function y_c.
4. Choose the trial form for y_p matching the type of $R(x)$.
5. Check if the trial duplicates a root of the auxiliary equation; find its multiplicity s.
6. Multiply the trial form by x^s if duplication occurs.
7. Write the particular solution y_p.
8. Write the GS $y = y_c + y_p$.
9. If initial conditions are given, evaluate the constants to write the solution of IVP.



Problem 1

- Solve $y''' - 6y'' + 11y' - 6y = e^{4x}$.
- Consider the HDE: $y''' - 6y'' + 11y' - 6y = 0$.
- The AE is $r^3 - 6r^2 + 11r - 6 = 0$.
- Roots of auxiliary equation: $1, 2, 3$ (all distinct).
- Thus, $y_c = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$.
- Since 4 is not a root, $s = 0$: trial $y_p = Ae^{4x}$.
- Substituting: $64A - 96A + 44A - 6A = 6A$.
- Setting $6A = 1 \Rightarrow A = \frac{1}{6}$. Thus, $y_p = \dfrac{1}{6} e^{4x}$.
- Therefore, the GS is given by $y = y_c + y_p$,

$$\boxed{y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x} + \frac{1}{6} e^{4x}}$$



Problem 2

- Solve $y''' - 6y'' + 11y' - 6y = e^{2x}$.
- Consider the HDE: $y''' - 6y'' + 11y' - 6y = 0$.
- The AE is $r^3 - 6r^2 + 11r - 6 = 0$
- Roots of AE are $1, 2, 3$. Thus, $y_c = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$.
- Since 2 is a simple root, $s = 1$: take $y_p = Axe^{2x}$.
- Substituting and simplifying:
  $y_p''' - 6y_p'' + 11y_p' - 6y_p = -Ae^{2x}$.
- Setting $-A = 1 \Rightarrow A = -1$.

  $y_p = -xe^{2x}$
- Therefore, the GS is given by $y = y_c + y_p$,

$$\boxed{y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x} - xe^{2x}}$$



Problem 3

- Solve $y''' - 3y'' + 3y' - y = e^x$.
- Consider the HDE: $y''' - 3y'' + 3y' - 1 = 0$.
- The AE is $r^3 - 3r^2 + 3r - 1 = 0$
- Auxiliary equation $(r-1)^3 = 0$.
- Thus, $y_c = (c_1 + c_2 x + c_3 x^2)e^x$.
- The root $r = 1$ with multiplicity 3, so $s = 3$: take $y_p = Ax^3 e^x$.
- Using $y = e^x u$: since $(D-1)^3(e^x u) = e^x u'''$, the equation reduces to $u''' = 1$.
- So $u_p = \dfrac{x^3}{6} \Rightarrow y_p = \dfrac{x^3}{6} e^x$. (Otherwise, do the direct computations)

$$\boxed{y = (c_1 + c_2 x + c_3 x^2)e^x + \frac{x^3}{6} e^x}$$



Problem 4

- Solve $y''' - y'' + y' - y = \sin x$.
- Consider the HDE: $y''' - y'' + y' - y = 0$.
- The AE is $r^3 - r^2 + r - 1 = 0$
- Roots: $1, \pm i$, and thus, $y_c = c_1 e^x + c_2 \cos x + c_3 \sin x$.
- Since $\sin x$ duplicates the $\pm i$ pair, $s = 1$: take $y_p = x(A \cos x + B \sin x)$.
- Substituting and collecting terms: $y_p''' - y_p'' + y_p' - y_p = (-2A - 2B)\cos x + (2A - 2B)\sin x$.
- Matching with $\sin x$: $-2A - 2B = 0$ and $2A - 2B = 1$.
- Solving: $A = \frac{1}{4}$, $B = -\frac{1}{4}$.
- Thus, $y_p = \frac{x}{4}(\cos x - \sin x)$.

$$\boxed{y = c_1 e^x + c_2 \cos x + c_3 \sin x + \frac{x}{4}(\cos x - \sin x)}$$

